INCH-POUND

MIL-DTL-2961E(SH)
30 April 2026
SUPERSEDING
MIL-V-2961D(SH)
3 April 1990

DETAIL SPECIFICATION

VALVES, GLOBE, PRESSURE REDUCING, GAS SERVICE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained globe (in-line and offset) pressure reducing valves for gas (air or nitrogen) service intended for shipboard and marine environment use (see 6.1).

1.2 Classification. Pressure reducing valves are of the following types, classes, body type, and pressure rating, as specified (see 6.2):

1.2.1 Types.

a. Type I – Direct spring loaded

b. Type II – Gas dome loaded

c. Type III – Gas dome loaded, pilot referenced

d. Type IV – Internal pilot operated

1.2.2 Classes.

a. Class 1 – Balanced main poppet

b. Class 2 – Unbalanced main poppet

1.2.3 Bodies.

a. Body A – In-line

b. Body B – Offset

1.2.4 Inlet pressure. Valves have nominal inlet pressure ratings of 200, 400, 600, 1,500, 3,000, 3,300, and 6,000 pounds per square inch (lb/in^2) (see 6.5.6).

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). The PIN to be used for pressure reducing valves acquired to this specification are created as follows:

M	**2961**	-	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
Military Specification prefix	Specification Number	-	Body Style (see code below)	Type (see code below)	Class (see code below)	Pressure rating (see code below)	Body material (see code below)	Valve inlet size (see code below)	Valve outlet size (see code below)	Connection compatibility (pipe or tube) (see code below)	Nominal valve size (see code below)

PIN Codes

Body style	**Code**
In-line	A
Offset	B

Type	**Code**
Direct spring loaded	1
Gas dome loaded	2
Gas dome loaded, pilot referenced	3
Internal pilot operated	4

Class	**Code**
Balanced main poppet	1
Unbalanced main poppet	2

Pressure rating	**Code**
200 lb/in²	02
400 lb/in²	04
600 lb/in²	06
1,500 lb/in²	15
3,000 lb/in²	30
3,300 lb/in²	33
6,000 lb/in²	60

Body material	**Code**
18-8 corrosion resistant steel	A
Aluminum bronze	B
Valve bronze	C
Naval brass	D
Copper-nickel-zinc or copper-nickel	E

End connections and nominal valve size	**Code**
0.25 inch	A
0.375 inch	B
0.50 inch	C
0.75 inch	D
1 inch	E
1.25 inches	F
1.5 inches	G
end connections compatible with pipe	P
end connections compatible with tube	T

PIN number example:

M2961-A1130CCDPC - Pressure reducing valve in accordance with MIL-DTL-2961 having an in-line body, direct spring loaded, balanced main poppet, 3,000 lb/in² rating, valve bronze body material, 0.5 inch inlet and 0.75 outlet connections compatible with pipe, 0.5 inch nominal valve size.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-H28/2	-	Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads- UN and UNR Thread Forms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-DTL-15024	-	Plates, Tags for Identification of Equipment, General Specification for
MIL-R-17131	-	Rods and Powders, Welding, Surfacing
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
DOD-F-24669	-	Forgings and Forging Stock, Steel Bars, Billets and Blooms, General Specification for (Metric)
DOD-F-24669/6	-	Forgings and Forging Stock, Steel Bars and Billets, Corrosion Resisting; for Reforging (Metric)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
MIL-STD-1474	-	Noise Limits
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-7063850	-	Tailpiece Assemblies Union End (BW SW SB) ¼ - 2 Inch 400 PSI WOG
810-1385884	-	Unions, Fittings and Adapters, Butt and Socket Welding, 6000 PSI, WOG-NPS

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919.)

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NAVSEA PUBLICATIONS

S9074-AQ-GIB-010/248 - Requirements for Welding and Brazing Procedure and Performance Qualification

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and repair for Machinery, Piping, and Pressure Vessels

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this publication are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection By Attributes

(Copies of these publications are available online at https://webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM A494/A494M - Standard Specification for Castings, Nickel and Nickel Alloy

ASTM A564/A564M - Standard Specification for Hot-Rolled, and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

ASTM B21/B21M - Standard Specification for Naval Brass Rod, Bar, and Shapes

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM B124/B124M - Standard Specification for Copper and Copper Alloy Forging Rod, Bar, and Shapes

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B151/B151M - Standard Specification for Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Rod and Bar

ASTM B169/B169M - Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar

ASTM B171/B171M - Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers

ASTM B283/B283M - Standard Specification for Copper and Copper-Alloy Die Forgings (Hot-Pressed)

(Copies of these publications are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE AMS-5362	-	Steel, Corrosion and Heat Resistant, Investment Castings 19Cr – 12Ni – 1.0Cb (Nb) Solution Heat Treated
SAE AMS-5363	-	Steel Castings, Sand and Centrifugal, Corrosion and Heat Resistant 18.5Cr – 10.5Ni – 0.90 (Cb + Ta) Solution Heat Treated
SAE-AMS-QQS763	-	Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings,
SAE J1926/1	-	Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of these publications are available online at www.sae.org.)

INSTRUMENT SOCIETY OF AUTOMATION (ISA)

ISA 75.01 - Control Valve Sizing Equations

(Copies of this publication are available online at https://www.isa.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be in accordance with 3.2.1 through 3.2.6. All materials shall be selected to prevent galling, seizing, or excessive wear between moving parts. However, this specification is not intended to be restrictive provided proposed alternative materials will give equal or better marine service than the materials specified. Proposed alternative materials shall be subject to approval by the contracting activity. Components for which specific materials are not specified shall be materials appropriate for the service intended. Particular attention shall be given to avoid sensitization of materials during hard facing, stress relieving, or welding. Use of cadmium plated parts and fasteners, including washers, is prohibited.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Body, gas dome or spring housing, and bottom cap. The body, gas dome or spring housing, and bottom cap shall be one of the following materials:

a. 303, 304, or 316; steel castings, sand and centrifugal, corrosion and heat resistant – SAE AMS-5363; steel castings, investment, corrosion and heat resistant – SAE AMS-5362; and steel, corrosion-resisting, bars and billets DOD-F-24669 and DOD-F-24669/6.

b. Aluminum bronze – grade 9BHT of ASTM B148, alloy 2 of ASTM B150/B150M, MIL-B-24480, or ASTM B124/B124M, ASTM B169/B169M, and ASTM B283/283M. When castings are used, the body shall be cast as a solid billet. Coring shall not be permitted.

c. Valve bronze – ASTM B61.

d. Naval brass – alloy B of ASTM B21/21M, or copper alloy numbers 462, 464, 482, and 485 of ASTM B21/21M, ASTM B124/B124M, and ASTM B283/283M unless prohibited by MIL-STD-777 or contractual limitations on body materials for new designs.

e. Copper-nickel-zinc alloy (nickel silver) or copper-nickel rod and bar, copper alloys UNS numbers C70600 and C71500 of ASTM B151/B151M. Copper alloy condenser tube plates, copper alloy UNS numbers C36500, C36600, C36700, C36800, C46400, C46500, C46600, C46700, C61300, C61400, and C71500 of ASTM B 171/B171M.

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3.2.3 Springs. Springs shall be made of phosphor-bronze, Monel or similar, corrosion resisting steel, or other corrosion-resistant metal.

3.2.4 Nonmetallic seating inserts, diaphragms, gaskets, O-rings, and similar parts. Nonmetallic parts shall be compatible with ship's air containing high moisture (100 percent humidity), entrained salt spray, and, on ships with conventional air compressors, MIL-PRF-17331 2190 TEP lubricating oil (see 6.2).

3.2.5 Valve poppets, seat rings, guide posts, and bushings. Internal trim shall be manufactured from corrosion-resistant alloy, DOD-F-24669 and DOD-F-24669/6 type 303, 303Se, 304, 316, 321, 347, or ASTM A564/A564M stainless steel; hardfacing material MIL-R-17131 (RCOCR-A or COCH-W), bronze; or nickel aluminum bronze. The guiding surfaces on the stem (guide posts) and the guide bushings shall have a minimum hardness differential of 50 Brinell hardness numbers. This requirement shall not apply if either the guide posts or bushings have a hardness in excess of 350 Brinell or are made of nongalling grade material (such as 303Se) and hardfacing material MIL-R-17131 (RCOCR-A or COCR-W). The finish on the guide post and bushings shall be not greater than 32 roughness height rating (RHR) in accordance with ANSI/ASME B46.1.

3.2.6 Bolting and other metallic parts not subject to line media. These parts shall be made from materials that provide inherent corrosion resistance in a marine atmosphere without requiring painting, coating, or plating. Selection of corrosion-resistant materials from MIL-DTL-1222 is recommended.

3.3 Construction. Construction of the valves shall be as specified in 3.3.1 through 3.3.9.

3.3.1 Concept. Internal trim shall be readily replaceable without requiring removal of the valve body from the line. Design emphasis shall be placed on simplicity, maintainability, ruggedness, and reliability, as defined in 3.3.1.1 through 3.3.1.4.

3.3.1.1 Simplicity. Valves shall have the fewest separate parts and fewest dynamic seals consistent with satisfactory performance. The valve shall require no adjustments during or after assembly, other than the set point adjustment (see 3.3.8).

3.3.1.2 Maintainability. Easy disassembly and reassembly with standard tools (see 3.6) shall be provided, and, as far as practical, the incorrect reassembly of parts shall be prevented. Positioning and alignment of parts in assembly shall employ positive means so that correct reassembly is repeatedly assured (see 3.3.5).

3.3.1.3 Ruggedness. To the maximum extent practical, valves shall tolerate and not be easily subject to damage, malfunction, or leakage resulting from foreign particle and other line media contamination or from mishandling. Therefore, valves shall not incorporate or rely upon excessively fine clearances or finishes. Valves shall tolerate occasional particulate contamination up to 40 microns in size.

3.3.1.4 Reliability. Valves shall operate satisfactorily for extended periods of time with minimum need for maintenance, lubrication, or adjustment, when subjected to the conditions normally associated with shipboard service. Valves shall not require maintenance or lubrication at intervals shorter than 1 year under continuous use in shipboard applications.

3.3.1.5 Welding. Welding shall be in accordance with S9074-AQ-GIB-010/248 and S9074-AR-GIB-010/278.

3.3.2 Pressure envelope. The nominal inlet (see 6.5.6), design (see 6.5.3), hydrostatic proof (see 6.5.7), and burst (see 6.5.2) pressures for the pressure containing envelope (body, gas dome or spring housing, and bottom cap) shall be as specified in table I. Design temperature (see 6.5.3) shall also be as specified in table I.

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TABLE I. Design and test pressures.

Nominal inlet pressure rating (lb/in²)	Design pressure (lb/in²)	Design temperature (°F)	Hydrostatic proof pressure (lb/in²)	Hydrostatic burst pressure (lb/in²)
200	240	165 max.	300	400
400	480	165 max.	600	800
600	720	165 max.	900	1,200
1,500	1,800	165 max.	2,250	3,000
3,000	3,600	165 max.	4,500	6,000
3,300	3,960	165 max.	4,950	6,600
6,000	7,200	165 max.	9,000	12,000
As required	Nominal inlet pressure × 1.2	165 max.	Nominal inlet pressure × 1.5	Nominal inlet pressure × 2.0

3.3.3 Seat construction. A gasket or O-ring sealed seat ring shall be provided that is either threaded in or retained by cage type construction. The poppet or plug guiding surfaces (bushings and posts) shall have the proper hardness (see 3.2.5), finish, concentricity, parallelism, clearances, length, and rigidity to prevent binding or seizing and to ensure proper seating under all operating conditions. These alignment requirements shall be maintained with interchangeable parts and under any tolerance stackup condition. A nonmetallic seating feature shall be incorporated in either the poppet or seat, and shall be readily replaceable on all sizes. A precision dimensioned gland or cavity shall be provided in either the body or seat ring to ensure proper and controlled retention of the sealing element.

3.3.4 Body construction. Valves shall be basic globe configuration with in-line or offset inlet and outlet ports. Pressure lines, including the reduced pressure sensing (impulse) line, gas dome loading line, and poppet balance line, where applicable, shall be internally ported in the valve body. Direct high velocity impingement of the flow issuing from the throttling orifice onto the sensing element shall be prevented. The bonnet or spring housing and bottom cap shall be attached to the body with bolted flanges or threaded union connection. Bearing surfaces of nuts and bolts and their respective mating surfaces on the valve shall be machine finished.

3.3.5 Interchangeability. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation and performance without requiring selection or fitting. In no case shall parts for a given valve be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength (see 6.5.4).

3.3.6 Springs. Springs shall not be compressed solid during any operation of the valve. Spring ends shall be squared and ground.

3.3.7 Threads. Threads shall conform to FED-STD-H28 and FED-STD-H28/2. Accidental loosening of threaded parts shall be prevented. External bolting shall be compatible with standard tools (wrenches).

3.3.8 Set point adjustment. The set point shall be adjustable through a range of at least 10 percent of the specified set pressure (see 6.2 and 6.5.9). The adjusting or loading device shall be safeguarded against accidental change in the set point. The option of a tamper-proof factory setting shall be available (see 6.2). The adjusting screw shall have a right-hand thread so that clockwise rotation increases the set pressure. The valve shall be adjustable while under pressure.

3.3.9 End connections. Both the inlet and outlet connections shall be designed for the maximum inlet pressure and shall be selected as shown in table II. Union thread pieces shall be cast or forged integral with the valve body or secured to the body with a straight thread and O-ring seal, with straight threads conforming to SAE J1926/1. The materials for the union end nut and tailpieces shall be as specified (see 6.2). The materials for the union threadpieces and union nuts shall be selected by the manufacturer from the materials specified herein to prevent galling. The following materials, in addition to those listed in the documents in table II, may be used for the union threadpieces and union nuts:

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a. ASTM A494/A494M for standard quality castings, MIL-C-24723 for high quality castings, H-Monel or similar

b. Corrosion-Resistant Stainless Steel (CRES), SAE AMS-QQ-S-763, composition 303Se or 416Se

The use of silver brazed tailpieces is prohibited in fire hardened spaces and applications. Unless otherwise specified (see 6.2), for additional information on materials selection for end connections see MIL-STD-777.

TABLE II. End connections.

Pressure rating (lb/in^2)	Applicable document
400	803-7063850 silver-brazed
1,500	810-1385884 silver-brazed
3,000	810-1385884 silver-brazed
6,000	810-1385884 weld end

3.4 Performance. Valves shall perform as specified in 3.4.1 through 3.4.9.

3.4.1 Springs. When removed from the sample valve and compressed fully (solid), the springs shall not have an excessive permanent set (see 4.6.2).

3.4.2 Hydrostatic burst. The valve pressure containing envelope (body, bonnet, dome, spring housing, bottom cap) shall withstand an internal hydrostatic burst pressure of two times the nominal inlet pressure (see 4.6.3 and 6.5.2).

3.4.3 Hydrostatic proof. The valve pressure containing envelope shall withstand an internal hydrostatic pressure of 1½ times the nominal inlet pressure (see 4.6.4 and 6.5.6).

3.4.4 Seat tightness. With nominal inlet pressure applied at the inlet and downstream pressure at lock up, the leakage rate from the inlet to outlet portion of the valve shall not exceed 15 standard cubic inches per hour (SCIH) per inch of seat diameter (see 4.6.5).

3.4.5 Set pressure limits (see 6.5.10). Set pressure shall be adjustable within the range or limits in 3.4.6 (see 4.6.6).

3.4.6 Accuracy of regulation (see 6.5.1). The valve shall meet accuracy of regulation requirements within the limits shown on figure 1 (see 4.6.7) or as otherwise specified (see 6.2).

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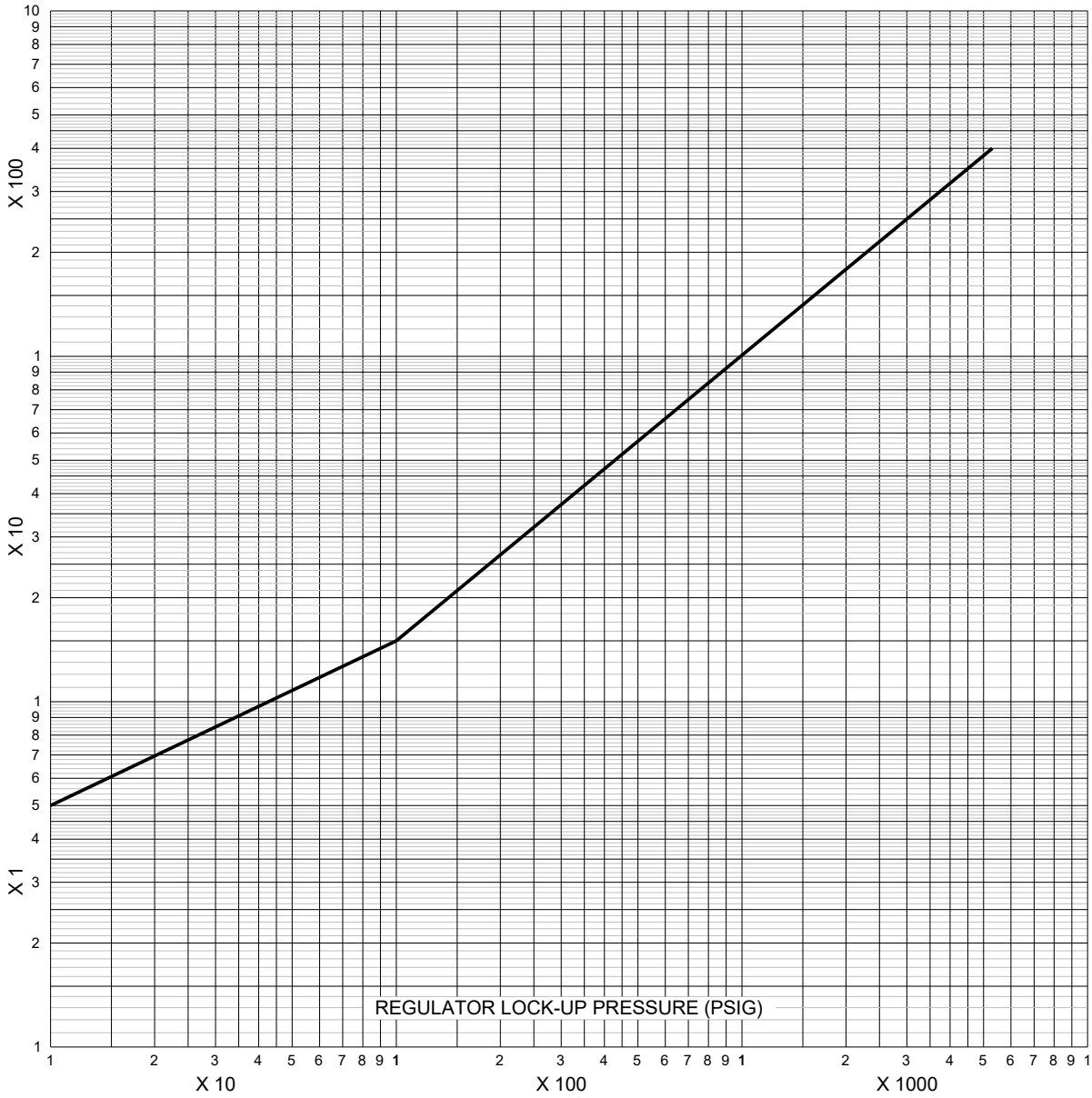


FIGURE 1. Accuracy of regulation.

3.4.7 Capacity requirements. Actual flow capacity in standard cubic feet per minute (SCFM) (see 6.5.11) shall be based on the minimum inlet pressure and highest reduced pressure setting under which the valve will be required to operate (see 6.2). The valve shall meet the specified capacity requirement, and shall operate without instability, hunting, and chattering, and within noise and vibration requirements. The valve shall also meet capacity requirements under all specified ambient operating conditions and within the requirements of 3.4.6 (see 4.6.8).

3.4.8 Noise and vibration.

3.4.8.1 Noise. Unless otherwise specified (see 6.2), when tested as specified (see 4.6.9.1) airborne and structureborne noise levels of the valve shall meet the MIL-STD-1474 and MIL-STD-740-2 levels for the "equipment grade" and "equipment type".

3.4.8.2 Vibration. The valve shall be in accordance with the requirements of 3.4.6 and 3.4.7 during environmental (type I) vibration in accordance with MIL-STD-167-1. There shall be no valve resonance frequencies below 33 hertz (Hz) (see 4.6.9.2).

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3.4.9 Shock test. The valve shall retain its set performance capability and suffer no structural damage or permanent deformation after shock testing in accordance with MIL-DTL-901 (see 4.6.10).

3.5 Marking.

3.5.1 Body markings. Valve bodies shall have the nominal inlet pressure rating, manufacturer's name or trademark, and flow direction arrow or the words "inlet" and "outlet" either cast or forged, stamped with round bottom dies, or laser etched on the valve body.

3.5.2 Identification plates. An identification plate in accordance with MIL-DTL-15024 and attaching device of corrosion-resisting steel, brass, or anodized aluminum shall be securely attached to the valve and shall list the following data or provide a space therefore:

a. Manufacturer's name or trademark.

b. PIN number (see 1.3).

c. Flow capacity rating of valve in SCFM at a given set of upstream and downstream pressure conditions or expressed as a flow coefficient (C_v).

d. Nominal inlet pressure rating.

e. Range of adjustable outlet pressures.

f. Manufacturer's drawing number.

g. Manufacturer's model and part number.

h. Applicable manual number.

i. Space for nine-digit Component Identification Data (CID) or allowance parts list (APL) number.

3.6 Special tools. Special tools shall not be required to install or service the valves. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.7 Workmanship. Components shall be free from defects that affect their appearance or which may affect their operation. Castings shall be clean, sound, smoothly cored (where permitted), true to form, uniform in texture, and free from cold shunts, porosity, blow holes, and other injurious defects. Castings shall not be plugged, brazed, or burned, nor otherwise repaired without prior approval of NAVSEA. Castings or bored billets shall be thoroughly cleaned, both inside and outside, with all fins, burrs, residue, and roughness removed. Machined surfaces shall not exceed 125 RHR. Welding and brazing shall be in accordance with 3.3.1.5.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed at a laboratory acceptable to the Technical Authority (6.5.12) on sample valves produced with equipment and procedures normally used in production. First article inspection shall consist of the examination and tests specified in table III.

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TABLE III. First article inspection.

Inspection	Requirements	Test method
Visual and dimensional examination	3.3.1 through 3.3.9 and 3.7	4.6.1
Tool (wrench) demonstration 1/	3.3.7	4.6.1
Spring test	3.4.1	4.6.2
Hydrostatic burst test	3.4.2	4.6.3
Hydrostatic proof test	3.4.3	4.6.4
Seat tightness test	3.4.4	4.6.5
Set pressure test	3.4.5	4.6.6
Accuracy of regulation test	3.4.6	4.6.7
Capacity requirement test	3.4.7	4.6.8
Noise, vibration, and shock test		
MIL-STD-167-1	3.4.8	4.6.9
MIL-STD-1474, MIL-STD-740-2	3.4.8	4.6.9
MIL-DTL-901	3.4.9	4.6.10

NOTE:

1/ The visual and dimensional inspection shall be required where specified in the test methods and at any point in the testing procedure where there exists a reasonable cause to examine a failure. The tool demonstration (see 3.3.7) may be accomplished at any opportune time. The interchangeability of spare parts (see 3.3.5) may be demonstrated in the tool demonstration.

4.2.1 Sampling for first article inspection. The number of valves to be subjected to first article inspection shall be two for each type, class, body, pressure rating, flow capacity, and reduced pressure range for which acceptance is desired.

4.2.2 Failures. Failure of any test shall be cause for failure of the first article inspection.

4.2.3 Order of tests. The order of tests shall be as shown in table III.

4.3 Conformance inspection. Conformance inspection shall consist of examination and tests specified in table IV.

TABLE IV. Conformance inspection.

Inspection	Requirements	Test method
Visual and dimensional examination	3.3.1 through 3.3.9 and 3.7	4.6.1
Hydrostatic proof test	3.4.3	4.6.4
Seat tightness test	3.4.4	4.6.5

4.3.1 Inspection lot. Valves of the same material, type, class, size, body design, and temperature rating ordered for delivery at the same time shall be considered a lot for purpose of sampling.

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4.3.2 Sampling for conformance inspection.

4.3.2.1 Sampling for visual and dimensional examination. A random sample shall be selected from each lot in accordance with MIL-STD-1916 or ANSI/ASQ Z1.4 for the visual and dimensional examination of 4.6.1 and table V.

TABLE V. Classification of defects.

Major	
101	Type, class, flow, size, capacity, material, pressure rating, range of discharge pressures, or temperature rating not as specified, or body configuration not a globe type.
102	End preparation not as required, or improperly machined.
103	Threads or bolting requirements not as specified.
104	Seat ring not properly retained.
105	Pressure envelope leaks.
106	Body passages conducive to fluid impingement or unreinforced areas.
107	Dimensional or structural change after hydrostatic test.
108	Dimensional errors, out of manufacturing tolerances.
109	Dirt, chips, sand, or residue from manufacturing processes present in valve.
110	Valve internal elements not concentric, parallel, square, or are out of round, galled, etc. Parallel disc/seat contact and free valve movement not evident.
111	Spring exhibits a “set” above tolerance or can be fully compressed under operation or adjustment. Spring ends not square or ground. Plating (where utilized) is cracked, delaminated, or otherwise defective.
112	Parts having the same manufacturer’s number not interchangeable without causing out of tolerance dimensional changes in assembly stack-up dimensions.
Minor	
201	Identification markings missing or illegible on valve.
202	Packaging/marking not as specified.
203	Tools (wrenches) not compatible with valve bolting.
204	Valve body must be removed from pipeline for repair.
205	Weight in excess by 10 percent or more of original estimate on submitted drawings.

4.3.2.2 Sampling for remaining conformance tests. A random sample shall be selected from each lot in accordance with table VI for the remaining tests specified in 4.3 and table IV.

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TABLE VI. Sampling for conformance, hydrostatic test, and seat tightness.

Allowable number lot size	Number of samples
2 to 8	Greater of 100% or 3
9 to 15	5
16 to 25	8
26 to 50	13
51 to 90	20
91 and over	32

4.3.3 Conformance tests. The sample items selected in accordance with 4.3.2.2 shall be subjected to the conformance tests shown in table IV. Any sample that fails any conformance test shall be rejected.

4.4 Disposition of sample units. When specified (see 6.2), sample units that have been subjected to conformance tests (see 4.3) and approved may be delivered in an assembled, ready-for-service condition.

4.5 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.6 and elsewhere herein.

4.5.1 Test temperature. Unless otherwise specified (see 6.2), inspection and tests shall be accomplished at ambient room temperature.

4.5.2 Test fluid. Unless otherwise specified (see 6.2), tests shall be conducted using clean fresh water. Sea water used as a test fluid shall have an approximate salinity of 35 parts per thousand.

4.6 Methods of inspection.

4.6.1 Visual and dimensional examinations. Each sample selected in accordance with 4.3.2.1 shall be weighed and visually and dimensionally examined to determine conformance with this specification and detailed engineering drawings. Major and minor defects shall be classified as shown in table V. Any sample failing to conform to the visual and dimensional examination shall be rejected.

4.6.2 Spring test. The spring from a disassembled sample valve shall be visually and dimensionally examined as follows:

a. The free spring length shall be measured and an allowance of 0.010 inch per inch of free spring length calculated. Fractional inches of free spring length shall be extrapolated and added to the calculation.

b. The spring shall be fully (solidly) compressed and released.

c. Ten minutes after release the spring shall be remeasured.

d. The spring shall not exhibit a permanent set in excess of the allowance calculated in step (a).

4.6.3 Hydrostatic burst test. One of the two sample valves (see 4.2 and 6.5.2) shall have its pressure envelope tested to the appropriate burst pressure in table I. Failure of the pressure envelope shall be that point at which the test pump cannot keep up with the leakage rate whether the deformation is permanent or temporary, or where there is an obvious structural failure, such as a split, crack, or rupture in the envelope. Once tested to the burst pressure, the tested valve envelope shall be disposed of in a manner precluding its subsequent repair or use even at lower pressure. A visual and dimensional examination shall be performed on valves that have not failed obviously as described above. An obvious failure below the burst pressure or a failure of the visual and dimensional examination shall constitute failure of this test.

4.6.4 Hydrostatic proof test. The hydrostatic proof test pressure shall be 1.5 times the nominal inlet pressure. Table I contains a listing of appropriate hydrostatic test pressures. The test shall be performed on the other sample valve. The valve shall be pressurized for 2 minutes, depressurized and pressurized again for 2 minutes. There shall be no leakage. At the completion of the test, the valve shall be visually and dimensionally examined in accordance with the visual and dimensional requirements of table III. Any leakage or a failure of the visual and dimensional examination shall constitute failure.

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4.6.5 Seat tightness test. The seat leakage rate after lockup shall be verified by measuring the pressure buildup in a known volume over a given period of time. The volume and time for this test shall be such that a pressure buildup of not more than 1 or 2 percent is required to indicate excessive leakage.

4.6.6 Set pressure limits. The set point limits shall be demonstrated in a manner or procedure similar to the seat tightness test specified in 4.6.5, at both the minimum and maximum limits of the set pressure range.

4.6.7 Accuracy of regulation. The valve shall be tested for accuracy of regulation, that is the amount the downstream regulated pressure can deviate from the set pressure at lockup and when the flow through the valve is increased to the rated capacity. An inlet pressure equal to the nominal rating shall be applied and flow varied from lockup to full flow and back to lockup. The pressure variation shall be within the limits of figure 1.

4.6.8 Capacity requirements. The valve shall be tested to the requirements of 3.4.7. Unless otherwise specified (see 6.2), testing and test calculations shall be in accordance with ISA 75.01 or other similar procedure acceptable to the Technical Authority.

4.6.9 Noise and vibration.

4.6.9.1 Noise. The valve shall be tested for airborne and structureborne noise in accordance with MIL-STD-1474 and MIL-STD-740-2. The valve shall be tested under the following operating conditions: maximum flow, maximum capacity, highest reduced pressure, mid-range pressure reduction, and lowest reduced pressure.

4.6.9.2 Vibration. The valve shall be tested for susceptibility to environmental vibration in accordance with MIL-STD-167-1 (see 3.4.8.2). Testing shall be accomplished with the nominal inlet pressure applied to the inlet port with the valve adjusted to the midpoint of the set pressure range.

4.6.10 Shock test. The valve shall be subjected to the high-impact mechanical shock requirements for grade A, class 1 of MIL-DTL-901 to determine its resistance to high impact mechanical shock. The shock test shall be performed with the nominal working pressure applied to the inlet port. During impact an instantaneous reversible pressure excursion is allowable. After shock the valve shall be tested in accordance with table IV and given a visual and dimensional check.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Reducing valves specified herein are intended for pressure reducing service in air or nitrogen systems in naval ships. They may be used in an Emergency Air Breathing (EAB) System where they are not the last valve in a series or cascade valve arrangement. This includes use of manual bypass valves and use of the priority side of an air system where appropriate. This specification does not include miniature or high precision pressure reducing valves generally associated with instrument or life support applications. Reducing valves procured to this specification are form, fit, and function replacements for prior revisions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. In line or offset globe body construction (see 1.2), type and class (when a specific type and class is required) (see 1.2).

c. Nominal size required (NPS) (see 1.2).

d. Pressure rating required (see 1.2).

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e. Whether a first article test is required (see 3.1).

f. Whether nonmetallic parts must be compatible with 2190 TEP (see 3.2.4).

g. Adjustable range of set pressures required and option of factory set tamper proof adjustment (see 3.3.8).

h. Material for union end nut and tailpiece (see 3.3.9).

i. Accuracy of regulation if other than required by figure 1 (see 3.4.6).

j. Flow capacity required (see 3.4.7).

k. When additional noise requirements must be met (see 3.4.8.1). Also specify "equipment grade" and "equipment type" (see MIL-STD-1474 and MIL-STD-740-2).

l. AQL of 0.65, inspection level III (see 4.3.2.2).

m. When sample units may be delivered in an assembled, ready-for-service condition (see 4.4).

n. If test conditions are other than specified (see 4.5, 4.5.1, and 4.5.2).

o. When alternate capacity requirements verification is acceptable (4.6.8).

p. Packaging requirements (see 5.1).

q. Requirements for Provisioning Technical Documentation (PTD), spare parts, and repair parts (see 6.3).

r. Engineering drawing technical content (see A.1.1).

6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts for the equipment covered by this specification should be as specified (see 6.2). When ordering spare or repair parts, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified. Provisioning of spare parts should meet special emphasis program requirements of the overall valve assembly. These programs included but are not limited to SUBSAFE, DSS-SOC, and MIC LVL 1.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Definitions.

6.5.1 Accuracy of regulation. The maximum permissible band over which the downstream pressure may vary when the valve is set at any pressure within the required range of adjustment and is subjected to any combination or inlet pressure, flow demand, and ambient temperature variations, within the specified limits.

6.5.2 Burst pressure. The maximum test pressure that a valve is required to withstand without complete or dangerous structural failure. The valve is not required to operate either during or following the application of burst pressure. Normally a burst test is not required under this specification, unless a new design has been submitted. If a burst test is specified, two first article samples are required.

6.5.3 Design pressure and temperature. The maximum pressure and temperature the valve should be subjected to under any condition. This is the pressure and temperature upon which the strength of the pressure containing envelope is based.

6.5.4 Interchangeability. Physical interchangeability in accordance with the following: interchangeable assemblies, components, and parts are those which are capable of being readily removed or replaced without alteration, misalignment, or damage to parts being installed or to adjoining parts. Interchangeable assemblies, components and parts also require no fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying, or forcing.

6.5.5 Lock-up pressure. The outlet pressure delivered by a reducing valve when the flow is reduced to a point where it is equal to or less than the allowable leakage.

6.5.6 Nominal pressure. The approximate maximum pressure that the valve will be subjected to in service under normal conditions.

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6.5.7 Proof pressure. The maximum test pressure that the valve is required to withstand without damage. Valve operation is not required during application of the proof pressure, but after the pressure has been removed, the valve must meet all performance requirements.

6.5.8 Sensitization. The changing of the structural or mechanical properties of a material.

6.5.9 Set pressure. The downstream pressure that the valve is set to maintain under a given set of operating conditions (that is, inlet pressure and flow). Normally the valve is set under conditions which will result in the highest downstream pressure (usually minimum inlet and lockup). Therefore, the delivered pressure would drop off or "droop" below the set pressure under all other operating conditions.

6.5.10 Set pressure limits. The range of set pressure over which the valve can be adjusted while meeting the performance requirements specified.

6.5.11 Standard air, conditions for SCFM. Standard Cubic Feet per Minute (SCFM) where standard conditions are temperature equals 68 °F, pressure equals 14.7 lb/in², and relative humidity equals 36 percent.

6.5.12 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Subject term (key word) listing.

Air systems
Dome
In-line
Nitrogen systems
Offset
Pilot
Poppet
Spring

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

A.1. SCOPE

A.1.1 Scope. This appendix covers the information that should be included on drawings when specified (see 6.2). This appendix is not a mandatory part of this specification. Information contained herein is for guidance only.

A.2. APPLICABLE PUBLICATIONS

A.2.1 General. The publications listed in this section are specified in appendix A of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in appendix A of this specification, whether or not they are listed.

A.2.2 Government publications.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-31000 - Technical Data Packages

(Copies of this publication are available online at https://quicksearch.dla.mil.)

A.2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3. DRAWINGS

A.3.1 Drawings. Drawings as specified in A.3.1.1 and A.3.1.2 should be prepared in accordance with MIL-STD-31000.

A.3.1.1 Preliminary drawings. Preliminary drawings should show the following:

a. An accurately scaled sectional assembly of the valve.

b. Bill of materials listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.

c. Complete details of all replaceable internal trim sufficient to permit manufacture, by ship or tender, of repair parts.

d. Dimensions and concentricity, parallelism, squareness, and roundness requirements necessary to evaluate the guiding alignment and running clearance limits of the main valve.

e. Finishes and hardnesses of the main valve dynamic guiding and sealing surfaces.

f. Installation dimensions.

g. End connection detail and clearance dimensions required for disassembly.

h. Required assembly torques or other equivalent assembly instructions for all threaded parts (except adjustments).

i. Table of spring data.

j. Estimated weight of complete valve assemblies for each combination of type, pressure, and size. For union end valves an estimate of nut and tailpiece weight should be stated separately.

k. Reference to any previous shock and vibration approval and test report numbers. Reference to previous burst test, if any, should be given.

l. Material descriptions where different from previously submitted preliminary drawings.

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A.3.1.2 Final drawings. In addition to the requirements of A.3.1 and A.3.1.1, the following data should be shown on final drawings:

a. Ship identification.
b. Applicable assembly drawing numbers.
c. Applicable manual number.
d. CID or APL number.
e. Application description.
f. Valve description.
g. The set pressure and adjustable range of valve.
h. Required accuracy of regulation over specified range of operating conditions.
i. Rated accuracy of regulation over specified range of operating conditions.
j. Required maximum capacity under specified conditions.
k. Rated maximum capacity under specified conditions.
l. Fail open capacity (for purposes of relief valve sizing).

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 4820-2026-001)

Review activity:
DLA – CC

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